

AERO40007 Mechanics 2022-23
Progress Test 2
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Question 1

A force $\underline{F} = 4\underline{i} + 5\underline{j} - 15\underline{k}$ passes through a point whose position vector is $\underline{R} = -2\underline{i} + 3\underline{j} - 5\underline{k}$. Find an equivalent system such that the force passes through a new point with position vector $\underline{R}_1 = 6\underline{i} - 4\underline{j} + 5\underline{k}$. **40%**

$$\underline{F}_1 = 4\underline{i} + 5\underline{j} - 15\underline{k}$$

$$\Delta \underline{R} = \underline{R} - \underline{R}_1 = -8\underline{i} + 7\underline{j} - 10\underline{k}$$

$$\therefore \underline{M}_1 = \Delta \underline{R} \times \underline{F}_1$$

$$= \begin{vmatrix} \underline{i} & \underline{j} & \underline{k} \\ -8 & 7 & -10 \\ 4 & 5 & -15 \end{vmatrix}$$

$$= (-105 + 55)\underline{i} + (-40 - 120)\underline{j} + (-40 - 28)\underline{k}$$

$$= -50\underline{i} - 160\underline{j} - 68\underline{k}$$

$$\therefore \text{Equivalent system is } \underline{F} = \underline{F}_1 + \underline{M}_1,$$

$$\text{where } \underline{F}_1 = 4\underline{i} + 5\underline{j} - 15\underline{k} \text{ and } \underline{M}_1 = -50\underline{i} - 160\underline{j} - 68\underline{k}$$

Question 2

What is the equivalent force system acting at point O in Figure 1.

60%

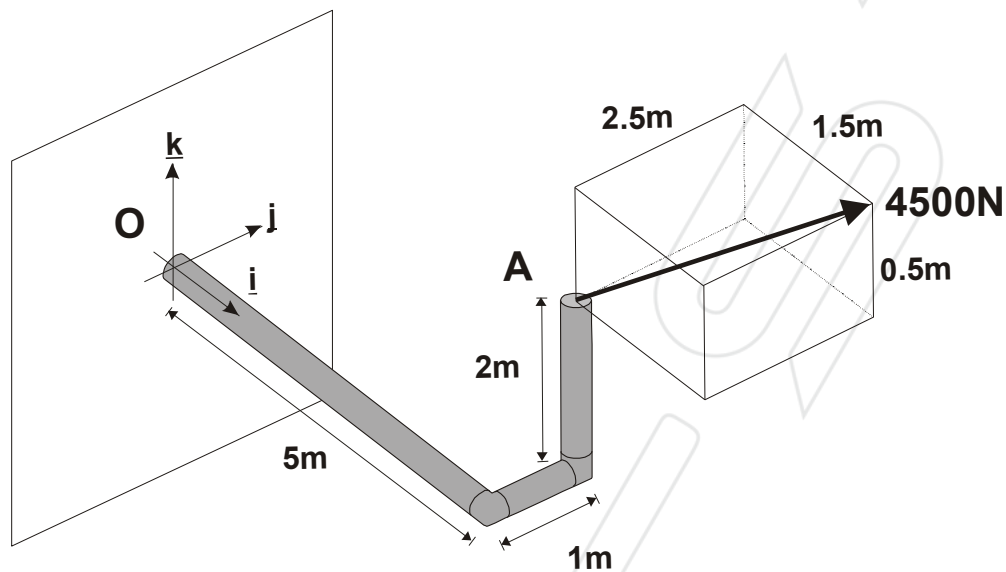


Figure 1

$$F_A = \left(\frac{1.5}{\sqrt{2.5^2 + 1.5^2 + 0.5^2}} \times 4500 \right) i + \left(\frac{2.5}{\sqrt{2.5^2 + 1.5^2 + 0.5^2}} \times 4500 \right) j + \left(\frac{0.5}{\sqrt{2.5^2 + 1.5^2 + 0.5^2}} \times 4500 \right) k$$

$$= 2281.92i + 3803.19j + 760.64k$$

$$\therefore F_O = F_A = 2281.92i + 3803.19j + 760.64k \text{ (N)}$$

$$\therefore \Delta R = R_A = 5i + j + 2k$$

$$\therefore M_O = \Delta R F_A$$

$$= \begin{vmatrix} i & j & k \\ 5 & 1 & 2 \\ 2281.92 & 3803.19 & 760.64 \end{vmatrix}$$

$$= -6845.74i + 760.64j + 16734.03k \text{ (Nm)}$$

$$\therefore (2281.92i + 3803.19j + 760.64k) \text{ N at A}$$

$$= (2281.92i + 3803.19j + 760.64k) \text{ N} + (-6845.74i + 760.64j + 16734.03k) \text{ N at O}$$